Software Quality Engineering

Assignment 2 - Student Course Registration System

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Project: Student Course Registration System

1. Black-Box Testing

Black-box testing was performed using Equivalence Class Partitioning (ECP), Boundary Value Analysis (BVA), and decision table testing. The tests checked credit limits, prerequisites, outstanding dues, course capacity, registration, and withdrawal conditions.

Black-box test results	Count
Total tests	44
Passed	35
Failed	9

Identified Defects

- Credit limit boundary condition does not allow exactly 18 credits.
- Prerequisite validation checks for any matching prerequisite instead of all required prerequisites.
- Outstanding dues are blocked only when greater than 1000 instead of greater than zero.
- Course capacity and waitlist logic do not correctly handle a full course.
- Withdrawal does not decrement the course enrollment count.

2. White-Box Testing

White-box testing examined the internal control flow of the program. Control Flow Graphs (CFG), cyclomatic complexity, independent paths, and Modified Condition/Decision Coverage (MC/DC) were considered.

Function	Cyclomatic Complexi
check_credit_limit	3
prerequisites_satisfied	2
registration_decision	8
register_course	4
withdraw_course	2
Total	19

Coverage Results

Statement coverage: 92.45% (49/53 statements).
Branch coverage: 80.77% (21/26 branches).
Combined pytest-cov result: 89%.

3. GitHub Collaboration and Automation

The project was maintained in a GitHub repository. A separate testing-improvements branch was used. Defects were documented through GitHub Issues, and changes were merged through Pull Requests. A GitHub Actions workflow was configured to install dependencies and execute automated tests.

4. Conclusion

The testing activities identified five main defect areas in the original registration system. Black-box testing checked expected behavior using input classes and boundaries, while white-box testing examined internal program paths and coverage. GitHub Issues, Pull Requests, and Actions provided a collaboration and automation workflow. The automated workflow currently reports failures because the original implementation contains the identified defects.